Bear EHR — ABA Billing Gap Analysis

NY Applied Behavior Analysis Policy Manual (Oct 2025) vs Bear EHR Current Capabilities

Prepared for	EMR-Bear Development Team
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1. Executive Summary

The Bear EHR has solid foundational billing infrastructure that can support ABA services. The billing pipeline (charges, claims, EOBs, X12, clearinghouses) is mature, and several critical ABA requirements — credential types, supervision relationships, NPI configurability, diagnosis-restricted fees, and billing policy enforcement — are already accommodated by the platform's existing flexibility.

This analysis is based on the eMedNY Applied Behavior Analysis Provider Policy Manual (effective October 1, 2025) which governs NY Medicaid fee-for-service and Managed Care billing for ABA services provided to members under 21 with an Autism Spectrum Disorder (ASD) or Rett Syndrome diagnosis.

> **Key finding:** Bear's architecture already covers the core billing mechanics for ABA. The `credential_types` table accommodates LBA/CBAA provider classifications. The `Supervision` model (with payor and location scoping) supports LBA supervisory chains. `BillingPolicy` / `PolicySet` can enforce ABA-specific procedure + diagnosis rules per plan. Fee-level `billing_level_id` restricts CPT codes to appropriate credential tiers. NPI routing is fully configurable via `Claim::FillClaim` and `billing_configuration.billing_mode`. The remaining gaps are: a supervision time/caseload compliance **report**, ABA-specific clinical **forms** (treatment plan, session notes), and billing **validation rules** (age, referral expiration, group size).

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2. What Bear Already Supports

ABA Requirement	Bear Capability	Status
CPT codes 97151–97158	`fees.procedure_code` accepts any 5-char code; `fees.minimum_time_to_bill` enforces 15-min threshold	Configure as fee schedule entries
Billing / Rendering / Supervising provider on claims	`billcase` and `charge` have `billing_provider_id`, `performing_provider_id`, `referring_provider_id`; claims have `f_17_provider_type`	Works today
LBA/CBAA credential types	`credential_types` lookup table exists — add LBA, CBAA, ABA Technician as records. `credentials` links user → credential_type → billing_level → payor	**Configure — no code changes**
Fee-to-credential-level restriction	`fees.billing_level_id` ties each CPT code to a credential tier. ABA fees can be restricted so only LBA- or CBAA-level providers generate charges	**Configure per fee**
Supervision relationships (LBA → CBAA/Tech)	`Supervision` model with HABTM `supervisors`, scoped by `payors` and `locations`. `users.clinically_supervised_by` provides direct supervisor chain. Validates no self-supervision or circular references	**Works today**
NPI routing per provider role	NPI fields on claims are highly configurable — `billing_configuration.billing_mode` (organization/provider/provider_location) controls NPI source; `Claim::FillClaim` populates f_* fields from provider credentials	**Works today**
Billing policy enforcement	`BillingPolicy` → `PolicySet` → `PolicyProcedure` + `ICD` combinations, scoped per plan. Can enforce ABA-specific procedure + diagnosis rules at the plan level	**Configure per plan**
Referral tracking	`xtreferrals` model with referrer, NPI, date, `provider_type` (referring / supervising)	Works today
Prior authorization with units	`authorizations` with `procedure_quantity`, `amount_cents`, `valid_from/until`, `fee_list` (comma-separated CPT codes), `remaining_units()`, `expired?()`	Works today

Diagnosis-restricted fees	`fees.specific_dxes` (HABTM with ICD codes) + `Procedure#validate_specific_dx()` enforces diagnosis match at billing time	Works today
15-minute unit billing	`fees.unit_type` supports "minutes", `fees.units` = 15	Configure per fee
Place of service	Full POS code table; `fees.place_of_service` stores comma-separated allowed POS codes per fee	Works today
Treatment plans	`tp/` module with goals, objectives, interventions, progress tracking, program-scoped assignments	Works today
Group sessions	`crew_teams` → `crew_sessions` → `crew_attendances` → per-client `encounter_forms` with individual superbills	Works today
Fee utilization limits	`fees.max_per_day`, `fees.max_per_year`, `fees.once_per_day`, `fees.minimum_time_to_bill`	Works today
ABA program enrollment	`programs` + `program_enrollments` — ABA as a program with client enrollment, insurance policy attachment, location scoping	Configure as new program

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3. Gaps That Need to Be Addressed

3.1 Supervision Time & Caseload Report — HIGH

While Bear's supervision system supports the LBA-to-CBAA/technician relationships, the ABA policy imposes specific **quantitative supervision requirements** that need a dedicated report to track and justify compliance.

Caseload limits:

- An LBA can supervise **no more than 6 individuals total** (any combination of CBAAs and unlicensed technicians)
- Examples: 1 CBAA + 5 technicians = 6 (at cap), 6 CBAAs + 0 technicians = 6 (at cap)

Time requirements per supervisee per calendar month:

- Supervision must be **5% minimum** of the hours the technician/CBAA spends providing ABA services
- At least **2 face-to-face, real-time contacts** (phone/email/text do not count)
- The LBA must **observe the supervisee providing services** in at least 1 of those contacts
- One of the two monthly contacts may occur in a small-group meeting

Data sources available in Bear for the report:

- **Supervisee service hours:** `encounter_forms` joined through `superbills` → `procedures` (units × 15 min) where `encounter_forms.user_id` = supervisee and `programs.id` = ABA program, grouped by calendar month
- **Supervisor identity:** `supervisions` table (HABTM `supervisors`) or `users.clinically_supervised_by`
- **Credential type:** `credentials` → `credential_types` per user — identifies LBA vs CBAA vs technician
- **Supervision encounters:** `encounter_forms` where `user_id` = LBA and `client_id` matches supervisee's caseload, filtered to supervision-relevant form types

- **Group vs individual contacts:** `crew_sessions` (group) vs direct encounter_forms (individual)
- **CPT 97155 qualification:** `charges.procedure_code = '97155'` on encounters where both LBA and technician are documented

What the report needs to show:

- Per LBA: current supervisee count (vs. 6 cap), with credential type breakdown (CBAA vs technician)
- Per supervisee per month: total ABA service hours delivered, total supervision hours received, supervision percentage (vs. 5% minimum)
- Per supervisee per month: count and type of face-to-face contacts (individual vs. group), observation flag
- Compliance status flags: at-risk (approaching thresholds) and non-compliant (below minimums)
- CPT 97155 sessions that qualify toward the 5% (only when LBA directs technician in a new/modified protocol)

> **Needed:** A supervision time and caseload compliance report that aggregates encounter data by LBA and supervisee, calculates supervision percentages, and flags non-compliance. The underlying data exists across `encounter_forms`, `supervisions`, `credentials`, and `crew_sessions` — the gap is the report aggregation and compliance dashboard, not the data capture. This is critical for audit readiness and for justifying the time and workload allocation of CBAAs.

3.2 Age Eligibility Enforcement — MEDIUM

ABA services under NY Medicaid are only covered for members **under 21 years of age**. The `people.date_of_birth` field is available on all clients, but there is no age-based eligibility check anywhere in the billing pipeline — not on `fees`, `procedures`, `superbills`, or `billcases`. The superbill-to-billcase conversion (`Superbill::BillcaseConverter`) validates insurance policy dates, authorization, and diagnosis, but does not check client age against any fee-level constraint.

> **Needed:** A `maximum_age` field on `fees` (or a validation in the superbill conversion flow) that rejects/flags billcases where the client's age at date of service is 21 or older. The check should compute age from `people.date_of_birth` against `billcase.date_of_service`.

3.3 Referral Validity Period Tracking — MEDIUM

ABA referrals are valid for **no more than 2 years**. The `xtreferrals` model has `referral_date` but no `expiration_date` or `valid_until` field. Note: while `authorizations` have robust expiration tracking (`valid_until`, `expired?()`, `about_to_expire?()`), the `xtreferrals` model has no equivalent — referral expiration is not enforced anywhere in the billing pipeline.

The referral must also capture specific information per policy:

- Age of the patient
- ASD or Rett Syndrome diagnosis
- Date of initial diagnosis
- Co-morbid diagnosis (if applicable)
- Symptom severity level / level of support
- Statement that the patient needs ABA services
- DSM-5 Diagnostic Checklist for ASD diagnoses

The referring provider must be a **NYS-licensed and NYS Medicaid-enrolled** physician, psychologist, psychiatric nurse practitioner, pediatric nurse practitioner, or physician assistant.

> **Needed:** Expiration date on referrals + validation that prevents billing ABA services against an expired referral + structured fields for the required referral content.

3.4 ABA-Specific Treatment Plan Form — MEDIUM

Bear has a robust treatment plan module, but ABA treatment plans require specific structured elements not present in the existing forms. See **Section 4** for the full detailed gap analysis.

3.5 Group Session Size Enforcement — LOW-MEDIUM

CPT codes 97154 and 97158 (group treatment) are limited to **max 8 individuals**. The `crew_teams` → `crew_memberships` → `crew_sessions` → `crew_attendances` pipeline creates per-client `encounter_forms` with individual superbills, which is correct for ABA group billing. However, there is no attendance cap validation — `crew_attendances` can be created for any number of clients per `crew_session`. CPT codes 97156 and 97157 (family guidance) have specific face-to-face requirements with guardians/caregivers not currently modeled.

> **Needed:** Validation on `crew_attendances` count per `crew_session` when the associated fee is 97154 or 97158, enforcing the 8-patient maximum. This could be a `before_create` callback on `Crew::Attendance` or a validation on `Crew::Session`.

3.6 Session Note / Progress Note Documentation — MEDIUM

Beyond the treatment plan, the policy (Section 8.3) requires clinical documentation for every ABA session that includes:

- Total hours of service per week
- Provision of services by LBA, CBAA, and/or technician (who delivered what)
- Location(s) of services (office, residence, community)
- Specific goals addressed and associated data to determine progress
- Data recorded using the collection method specified in the treatment plan

Bear's encounter forms can capture session notes, but there is no ABA-specific session note template that enforces structured data collection tied to treatment plan goals.

> **Needed:** An ABA session note form/page type that links to the client's active ABA treatment plan goals and requires structured data entry per goal per session.

3.7 MMC vs FFS Billing Distinction — LOW-MEDIUM

ABA services were carved into the Medicaid Managed Care (MMC) benefit package effective **January 1, 2023**. This means:

- For **FFS members**: bill NYS Medicaid directly using the ABA fee schedule
- For **MMC members**: bill the member's specific managed care plan, which may have its own rates, prior authorization requirements, and billing procedures

Bear already supports multiple payors and plans on `insurance_policies`, but there is no ABA-specific logic to route claims differently based on FFS vs MMC status or to flag when an MMC member's plan should be contacted for coverage/billing guidance.

> **Needed:** Awareness-level enhancement — ensure ABA billing workflows check the client's insurance type and surface appropriate guidance for MMC members.

3.8 Record Retention Policy Flag — LOW

ABA records must be kept for a **minimum of 6 years** and, for minors, **until the patient turns 22 years of age** (whichever is later). Bear does not currently have a service-specific retention policy flag.

> **Needed:** A retention policy attribute on ABA-related records (or program-level configuration) that prevents premature archival/purging.

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4. ABA Treatment Plan Form — Detailed Gap Analysis

4.1 Bear's Current Treatment Plan Structure

Bear has a sophisticated TP system with a reusable hierarchy:

```
Problem (ICD-linked)
  ├── Goals
  │     ├── Objectives (state machine: new → met/partially_met/not_met/deferred/
  │     └── Interventions
  └── Symptoms/Risks (with assessment levels)
```

Per-client assignments track progress via `Outcome` records, support due dates, program-based scoping, and life domain mapping. The `assigned_objectives` have quantitative from/to tracking with configurable unit types.

There is also an existing `AppliedBehaviorResult` page model — but it is minimal, only capturing an ABA yes/no flag, referral info, provider names/NPIs, and a diagnosis code. This is a **referral documentation page**, not a treatment plan.

4.2 Gaps by Section

A. Behavioral Assessment Section — No Current Equivalent

The ABA treatment plan must document the **assessment methodology and results** before any goals are set. The assessment itself is a billable service (CPT 97151).

Required Field	Bear Equivalent	Gap
Assessment methodology (ABC log, FBA, behavioral observation, self-monitoring, inventory)	None — TP starts at Problem assignment	New section needed
Results of assessment	None	New field
Standardized assessments used (adaptive behavior scales, symptom inventories, aggression ratings)	None	New field
Results of standardized assessment	None	New field

B. Operationally Defined Target Behaviors — Partially Mapped

Required Field	Bear Equivalent	Gap
Operationally defined target behavior (increase or decrease)	`assigned_goal.client_goal` (free text)	Needs structured direction: increase / decrease / maintain
Baseline data in objective terms (frequency, intensity, duration)	`assigned_objective.from_quantity` + `from_type`	Partially works — unit types limited
Current data in same method as baseline	`outcome.current_quantity` + `current_type`	Same limitation on types
Date of goal introduction	`assigned_goal.created_at`	Works but not a user-facing field
Sampling method for ongoing data collection	None	New field needed
Expected frequency of data collection	None	New field needed

Bear's `objective.default_unit_type` options are: *1-10 scale, percent, times per day/week/month, episode*. ABA needs more granular behavioral measurement types: **rate (per hour), duration (minutes), latency (seconds), interval recording percentage, frequency count**.

C. Mastery & Generalization Criteria — Not Present

Required Field	Bear Equivalent	Gap
Criterion for mastery per goal	`assigned_objective.to_quantity` (single target number)	New structured fields needed
Maintenance criteria	None	New field
Generalization criteria	None	New field
Conditions/context for performance	None	New field
Instructional procedure per goal	`assigned_intervention` (named, not described)	Needs structured procedure description

D. Background & Referral Context — Partially Mapped

Required Field	Bear Equivalent	Gap
Reason for referral	`xtreferral.source_note`	Could work
Background info (psychosocial, family, educational, medical history)	Scattered across intake forms	New section or cross-reference
Client strengths	None on TP	New field
Current educational/therapeutic services	None on TP	New field
Response to current services & barriers	None on TP	New field
Treatment setting	POS on charge, not on TP	New field on TP
Plan for coordination with other providers	None	New field
Plans for transition and discharge	`discharge_review` exists but is retrospective	New prospective section

E. Caregiver/Family Training Goals — Not Present

Required Field	Bear Equivalent	Gap
Parent/caregiver training goals	None — goals are client-focused	New goal type or flag
Generalization of skills to home/community	None	Part of caregiver training section

This maps to CPT 97156 (family guidance) and 97157 (multi-family guidance). The treatment plan should explicitly identify which goals involve caregiver training so billing for these codes can be linked.

F. 6-Month Mandatory Review Cycle — Partially Mapped

Required Field	Bear Equivalent	Gap
TP updated every 6 months minimum	`encounter_form.next_due_date` + form recurrence	Can be configured but needs enforcement
Updated goals and data	`goal_review` model exists	Works
Relevant changes in history	None structured	New field on review
Shared with referring provider	None tracked	New field (date shared)

The policy also requires that initial treatment plans and updates be **shared with the referring provider** and, as necessary, other care providers (primary care, PT, OT, speech).

4.3 Proposed ABA Treatment Plan Form Structure

Section 1: Client Background

- Reason for referral (text)
- Referral date + referring provider (pull from `xtreferral`)
- ASD/Rett diagnosis with date of initial diagnosis (pull from `diagnoses`)
- Severity level / level of support (dropdown: Level 1 / Level 2 / Level 3)
- Comorbid diagnoses (multi-select from client's dx history)
- Client strengths (text)
- Psychosocial / family / educational / medical history summary (text)
- Current services being received (text)
- Response to current services and barriers to treatment (text)

Section 2: Behavioral Assessment

- Assessment date(s)
- Assessment methodology (multi-select: ABC log, functional behavior assessment, behavioral observation/sampling, self-monitoring/self-report, inventory, other)
- Assessment methodology description (text per method)
- Standardized assessments used (multi-entry: assessment name + score + date)
- Assessment results summary (text)

Section 3: Target Behaviors & Goals

For each target behavior (extends Bear's `assigned_problem` → `assigned_goal`):

- Operationally defined target behavior (text)
- Direction: increase / decrease / maintain (enum)
- Baseline data: value + unit (frequency / rate / duration / latency / interval %)
- Date of introduction
- Data collection method (dropdown: frequency count, rate recording, duration recording, latency recording, interval recording, time sampling, permanent product)

- Expected collection frequency (e.g., "every session", "3x/week")
- Current data: value + unit (same measurement as baseline)
- Mastery criterion (text, e.g., "80% across 3 consecutive sessions with 2 different therapists")
- Maintenance criterion (text)
- Generalization criterion (text)
- Instructional procedure (text — the teaching method)
- Conditions/context for performance (text)

Section 4: Caregiver Training Goals

- Caregiver training goal (text)
- Target skills for caregiver (text)
- Generalization plan (text)
- Links to CPT 97156/97157 billing

Section 5: Service Delivery Plan

- Recommended hours per week (numeric)
- Breakdown by provider type: LBA direct hours, CBAA hours, technician hours
- Treatment setting(s) (multi-select: home, clinic, community, school — noting school is not Medicaid-reimbursable)
- Estimated duration of treatment (months)

Section 6: Coordination & Discharge

- Other providers involved (multi-entry: name, role, contact)
- Coordination plan (text)
- Transition criteria (text)
- Discharge criteria (text)

Section 7: Review (Every 6 Months)

- Review date

- Updated goals and data (per goal: current data, status change, goal modifications)
- Changes in psychosocial / family / educational / medical history
- Reviewer signature (LBA)
- Shared with referring provider (date)

4.4 Implementation Approach

Option A: New page type — Create an `aba_treatment_plan` page model (like `applied_behavior_result` but much richer) that lives alongside the existing TP. Sections 1–2 and 4–7 are new fields; Section 3 extends the existing `assigned_goal` / `assigned_objective` system with ABA-specific columns.

Option B: Extend existing TP — Add ABA-specific fields to `assigned_goals`, `assigned_objectives`, and `treatment_plans` (with flags to show/hide based on program type). This keeps everything in one TP system but adds complexity to the existing code.

> **Recommendation:** Option A is cleaner. The existing TP system is behavioral-health-generic and already complex (17 models under `tp/`). An ABA-specific page would let you build the exact form the policy requires without retrofitting the general TP structure.

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5. ABA CPT Codes Reference

CPT Code	Description	Unit	Who Delivers	Who Bills
97151	Behavior identification assessment — administered by LBA, face-to-face with patient and/or caregiver, including non-face-to-face analysis and report preparation	15 min	LBA	LBA
97152	Behavior identification-supporting assessment — administered by technician under LBA direction, face-to-face with patient	15 min	Technician	LBA
97153	Adaptive behavior treatment by protocol — administered by technician under LBA direction, face-to-face, one patient	15 min	Technician	LBA
97154	Group adaptive behavior treatment by protocol — administered by technician under LBA direction, face-to-face, **2+ patients (max 8)**	15 min	Technician	LBA
97155	Adaptive behavior treatment with protocol modification — administered by LBA, may include simultaneous technician direction, face-to-face, one patient	15 min	LBA	LBA
97156	Family adaptive behavior treatment guidance — administered by LBA, face-to-face with guardian(s)/caregiver(s), **with or without patient present**	15 min	LBA	LBA
97157	Multiple-family group adaptive behavior treatment guidance — administered by LBA, face-to-face with **multiple sets of guardians/caregivers, without patient**	15 min	LBA	LBA
97158	Group adaptive behavior treatment with protocol modification — administered by LBA, face-to-face, **multiple patients (max 8)**	15 min	LBA	LBA

Billing Rules:

- **97156 & 97157:** Can only be billed when the service is delivered as part of the child's treatment plan
- **97154 & 97158:** Can only be billed for group sessions of **no more than 8 individuals**
- **97155 & the 5% rule:** May count toward the 5% supervision minimum **only** when the LBA joins the patient and technician during a treatment session to direct the technician in implementing a **new or modified treatment protocol**
- All codes are billed in **15-minute increments**
- All ABA services require an **ASD or Rett Syndrome diagnosis** (DSM-5) and the client must be **under 21 years of age**

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6. ABA NPI Routing Reference

Bear's NPI fields are highly configurable and can accommodate all ABA provider role scenarios. The table below documents the correct NPI mapping for reference during configuration:

Service Delivered By	Billing NPI	Supervising NPI	Rendering NPI
LBA directly	LBA	—	LBA
CBAA (supervised by LBA)	Supervising LBA	Supervising LBA	CBAA
Unlicensed technician	Supervising LBA	Supervising LBA	Supervising LBA
LBA limited permit holder	Supervising LBA	Supervising LBA	Supervising LBA

Key rules:

- For CBAAs, the CBAA's own NPI appears **only** in the Rendering Provider field. Billing and Supervising always use the supervising LBA's NPI.
- For unlicensed technicians and LBA limited permit holders, **all three NPI fields** use the supervising LBA's NPI.
- CBAAs cannot bill Medicaid directly — the supervising LBA's NPI must be used for billing.

- LBAs can bill for services provided by non-enrolled unlicensed aides in their multi-disciplinary team.

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7. Priority Implementation Order

Priority	Item	Type	Rationale
1	ABA fee schedule setup (97151–97158, ICD restrictions, 15-min units)	Configuration	Core billing — can proceed immediately, no code changes
2	LBA/CBAA/Technician credential types	Configuration	Add credential types to accommodate ABA provider roles
3	Supervision relationships (LBA → supervisees)	Configuration	Set up LBA-to-CBAA and LBA-to-technician supervisory assignments
4	NPI routing per provider role	Configuration	Configure billing/rendering/supervising NPIs per the routing table in Section 6
5	Supervision time & caseload compliance report	**New development**	Critical for audit readiness — justifies CBAA time/load and tracks 5%/2-contact minimums
6	Referral expiration tracking (2-year validity + required fields)	New development	Billing prerequisite — services without valid referral are not covered
7	Age eligibility validation (under 21)	New development	Prevents billing for ineligible clients
8	ABA treatment plan form with structured fields	New development	Clinical documentation requirement; claims may be audited
9	ABA session note template	New development	Per-session documentation linked to treatment plan goals
10	Group session size enforcement (max 8 for 97154/97158)	New development	Edge case validation
11	MMC vs FFS routing awareness	Enhancement	Guidance-level enhancement
12	Record retention policy flag	Enhancement	Low urgency; administrative

> **Note:** Items 1–4 are configuration tasks that can proceed immediately with no code changes. Item 5 (supervision report) is the highest-priority new development, as it is required to demonstrate compliance during audits and to justify the time allocation and caseload of CBAAs working under LBA supervision.

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8. Next Steps

1. **Configure fee schedules** — Set up ABA fee schedule with CPT 97151–97158, attach ASD/Rett ICD restrictions via `specific_dxes`, configure 15-minute units. This can start immediately.

2. **Configure credential types and supervision** — Add LBA, CBAA, and ABA Technician credential types. Set up supervisory relationships in the existing supervision system. Configure NPI routing per the reference table in Section 6.

3. **Build supervision time & caseload report** — This is the first required development item. The report should aggregate encounter data by LBA and supervisee, calculate supervision percentages against service hours, track face-to-face contact counts, and flag non-compliance. This is essential for justifying CBAA workload and audit readiness.

4. **Build ABA treatment plan form** — Design the page type per Section 4.3, decide on Option A vs B, and build incrementally (background section first, then goals, then reviews).

5. **Validate with stakeholders** — Confirm which Bear agencies will be providing ABA services and whether they bill FFS, MMC, or both.

6. **Pilot and iterate** — Start with a single agency, validate claim acceptance with eMedNY / clearinghouse, and refine configuration based on real rejection data.